

# Formal Verification of the Heat Equation: Formulation, Discretization, and Conditioning in Lean 4 with an Application to Compression Operators

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## Abstract

We develop, and partially formalize in Lean 4, the standard verification pipeline for a model linear parabolic PDE: weak formulation and well-posedness via Lax–Milgram (Part I), finite-difference discretization with consistency and stability (Part II), and spectral conditioning analysis of the resulting linear systems (Part III). None of the underlying mathematics is new; the contribution at this stage is an honest, machine-checked scaffold with every open obligation explicitly tracked. Part IV synthesizes the pipeline and gives a full sorry/axiom ledger. Part V is the paper’s one genuinely new observation: the passage from a continuous function to its finite mesh representation is dimension-reducing ( $\dim(X_C) < \dim(X)$ ) but *fails* the non-collapse condition (Assumption 2.3) that Principia Orthogona, Volume I, Definition 3.1 requires of a compression operator — and this failure is exactly its non-injectivity. It is therefore a *non-feasible* compression in Volume I’s precise sense, and that failure of feasibility is precisely what makes the reverse (discrete-to-continuous) direction underdetermined — a fact independent of, and structurally different from, the heat equation’s own time-irreversibility (its  $H$ -theorem). We prove both facts explicitly and keep them separate throughout, since conflating them would overstate what either one shows.

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## Introduction: motivation and honest scope

**Why verify numerical PDE solvers, not just run them.** A finite difference or finite element code that appears to converge in practice can still harbor a discretization or conditioning bug that only manifests on meshes or right-hand sides not covered by testing. Formal verification closes this gap for the mathematical content of the method (consistency, stability, conditioning bounds) independently of any particular implementation. This is now an active area: large-scale Lean formalizations of elliptic regularity theory (2026) demonstrate the ecosystem can support proofs at this level of analytic sophistication.

**Scope: one model problem, carried end to end.** We fix the simplest genuinely instructive case — the 1D heat equation with homogeneous Dirichlet boundary conditions — and follow it through weak formulation, discretization, and conditioning, rather than surveying many methods shallowly. Every step is either proved in Lean, given a precise `sorry`-tagged statement with a stated proof route, or explicitly marked as prose-only.

**What “verified” means here — and what it does not.** At the time of this draft, **no file has been compiled against Mathlib4**. Every declaration below should be read as a design sketch: the mathematical content has been checked by hand (and, where stated, independently re-derived symbolically or numerically in this session), but lemma and API names against the actual Mathlib API are best-effort and unverified. The full `sorry`/axiom ledger is given in Part IV, §18, and is not hand-maintained prose — it is generated directly from the `sorry` count in the source files.

**What this monograph contributes, precisely.** Parts I–III formalize standard, well-known results (Lax–Milgram well-posedness, the Lax equivalence theorem, tridiagonal conditioning) — the contribution there is the Lean scaffold itself, not new mathematics. Part V contains the one new observation: discretization is literally an instance of the Compression operator already defined and used throughout the Principia Orthogona series, and the series’ own structural fact about compression operators (dimension-reducing  $\Rightarrow$  generically non-injective) explains, precisely and without overreach, why the *reverse* passage — discrete data back to a continuous function — is not merely difficult but formally underdetermined. We are explicit throughout Part V about what this does *not* show: it does not show the heat equation is a  $\text{dm}^3$  system, has a fold, or exhibits any bifurcation structure. A direct check (the discretized backward-Euler operator’s eigenvalues, computed for a representative mesh) confirms the discretized dynamics are a simple global linear contraction to a single fixed point, for every mesh size and time step — the opposite of fold/threshold structure. The contribution of Part V is narrow and precisely scoped: one operator,  $C$ , gains one new, independent, classical instance outside the framework’s original setting.

## Part I

# Continuous Formulation

## 1 Function-Space Preliminaries

Let  $\Omega = (0, 1)$ . We work in  $L^2(\Omega)$  and its subspace  $H_0^1(\Omega)$ , the closure of  $C_c^\infty(\Omega)$  in the  $H^1$  norm — functions with one weak derivative in  $L^2$  and zero trace at the boundary.  $H_0^1(\Omega)$  is the natural setting for homogeneous Dirichlet problems: it builds the boundary condition into the function space itself, so the weak formulation below needs no separate boundary term.

*Remark 1.1* (Lean status). A full Lean construction of  $H_0^1(0, 1)$  as a complete inner product space is nontrivial and is its own formalization milestone: Mathlib’s `MeasureTheory` and `Analysis.Calculus.FDeriv` machinery supplies weak derivatives, but assembling them into a complete Hilbert space with the right norm requires real work. In the current scaffold this is recorded as an axiom-backed placeholder (`H10` in `HeatEquation_Step1.lean`) so that downstream statements typecheck; building the real object is the concrete next step and, as the source file itself notes, is the genuinely hard part of Part I — everything after it is comparatively mechanical.

## 2 The Weak Formulation

The strong form of the model problem is

$$u_t = u_{xx} \text{ on } (0, 1) \times (0, T], \quad u(0, t) = u(1, t) = 0, \quad u(x, 0) = u_0(x).$$

One backward-Euler step of size  $\tau$  reduces this to an elliptic boundary value problem for the new time level  $w$ :

$$w - \tau w'' = u_{\text{prev}} \text{ on } (0, 1), \quad w(0) = w(1) = 0.$$

Multiplying by a test function  $v \in H_0^1$  and integrating by parts (the boundary term vanishes by  $v \in H_0^1$ ) gives the weak formulation: find  $w \in H_0^1$  such that

$$a_\tau(w, v) := \int_0^1 w' v' dx + \tau^{-1} \int_0^1 w v dx = \tau^{-1} \int_0^1 u_{\text{prev}} v dx = \langle f, v \rangle \quad \forall v \in H_0^1.$$

*Remark 2.1* (Lean status). The bilinear form  $a_\tau$  is recorded abstractly in the current scaffold (`a` in `HeatEquation.Step1.lean`) rather than as the literal integral above, since the literal form requires the  $H_0^1$  construction of §1. Once that exists,  $a_\tau$  should be replaced by the explicit integral expression.

### 3 Well-Posedness via Lax–Milgram

**Proposition 3.1** (Coercivity). *For  $\tau > 0$ ,  $a_\tau$  is coercive on  $H_0^1(0, 1)$ : there exists  $\alpha > 0$  with  $a_\tau(w, w) \geq \alpha \|w\|_{H_0^1}^2$  for all  $w \in H_0^1$ .*

*Proof sketch.*  $a_\tau(w, w) = \|w'\|_{L^2}^2 + \tau^{-1} \|w\|_{L^2}^2 \geq \min(1, \tau^{-1}) \|w\|_{H^1}^2$ , using that the  $H^1$  norm is exactly  $(\|w'\|_{L^2}^2 + \|w\|_{L^2}^2)^{1/2}$ . Both terms in  $a_\tau(w, w)$  are already nonnegative sums of squares, so coercivity here needs no Poincaré inequality — the mass term  $\tau^{-1} \|w\|_{L^2}^2$  alone already controls the  $L^2$  part, and the stiffness term controls the derivative part directly.  $\square$

**Proposition 3.2** (Boundedness). *There exists  $C_\tau > 0$  with  $|a_\tau(w, v)| \leq C_\tau \|w\|_{H_0^1} \|v\|_{H_0^1}$  for all  $w, v \in H_0^1$ .*

*Proof sketch.* Cauchy–Schwarz on each of the two integrals:  $|\int w' v'| \leq \|w'\|_{L^2} \|v'\|_{L^2}$  and  $\tau^{-1} |\int w v| \leq \tau^{-1} \|w\|_{L^2} \|v\|_{L^2}$ ; sum and bound by  $C_\tau = \max(1, \tau^{-1})$  times the product of  $H^1$  norms.  $\square$

**Theorem 3.3** (Well-posedness of one backward-Euler step). *For every  $\tau > 0$  and every  $f \in H_0^1(0, 1)^*$  (represented via Riesz as an element of  $H_0^1$ ), there exists a unique  $w \in H_0^1(0, 1)$  with  $a_\tau(w, v) = \langle f, v \rangle$  for all  $v \in H_0^1$ .*

*Proof.* Immediate from Propositions 3.1–3.2 and the Lax–Milgram theorem.  $\square$

*Remark 3.4* (Lean status). Propositions 3.1–3.2 and Theorem 3.3 are stated with `sorry` in `HeatEquation.Step1.lean` (`a_coercive`, `a_bounded`, `weak_solution_exists_unique`). Once `H10` and `a` carry real definitions, the coercivity and boundedness proofs above are short computations; the remaining risk is purely the exact Mathlib API surface for `IsCoercive/LaxMilgram`, which was written from best recollection and must be checked against the installed Mathlib version.

## 4 Chapter Summary and Formal Status Table

| Statement                  | Status | Notes                                                              |
|----------------------------|--------|--------------------------------------------------------------------|
| $H_0^1(0, 1)$ construction | axiom  | Placeholder; real construction is the key remaining task of Part I |
| Weak form $a_\tau$         | axiom  | Abstract stand-in for the literal integral                         |
| Coercivity (Prop. 3.1)     | sorry  | Short proof once $H_0^1$ , $a_\tau$ are real                       |
| Boundedness (Prop. 3.2)    | sorry  | Cauchy–Schwarz, mechanical                                         |
| Well-posedness (Thm. 3.3)  | sorry  | Depends on exact Lax–Milgram API                                   |

## Part II

# Discretization

## 5 From Continuous to Discrete

Fix  $N \in \mathbb{N}$  and mesh spacing  $h = 1/(N + 1)$ , with grid points  $x_i = ih$ ,  $i = 0, \dots, N + 1$ . Approximate  $u''(x_i)$  by the central difference

$$u''(x_i) \approx \frac{u_{i-1} - 2u_i + u_{i+1}}{h^2}.$$

## 6 The Discrete Operator as a Matrix

The resulting operator is the symmetric tridiagonal matrix

$$A_h = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 \end{pmatrix} \in \mathbb{R}^{N \times N},$$

and the discrete bilinear form is the quadratic form  $w \mapsto w^\top A_h w$  on  $\mathbb{R}^N$ , the direct discrete analogue of  $\int (w')^2$ .

## 7 Consistency

**Proposition 7.1** (Truncation error). *For  $u \in C^4$ , the central difference approximates  $u''(x_i)$  with error  $O(h^2)$ : writing  $\tau_i$  for the truncation error,*

$$\tau_i = \frac{u(x_{i-1}) - 2u(x_i) + u(x_{i+1}))}{h^2} - u''(x_i) = \frac{h^2}{12} u^{(4)}(\xi_i)$$

for some  $\xi_i \in (x_{i-1}, x_{i+1})$ .

*Proof sketch.* Taylor-expand  $u(x_i \pm h)$  to fourth order and combine; the third-order terms cancel by symmetry, leaving the stated  $O(h^2)$  remainder via the Lagrange form of the Taylor remainder.  $\square$

*Remark 7.2* (Lean formalization point). This is a quantitative Taylor-remainder bound — Mathlib’s `taylor_mean_remainder` family (exact name TBC) should supply the remainder term directly; assembling the two-sided combination and the constant  $1/12$  is the remaining work.

## 8 Stability

**Proposition 8.1** (Energy stability of backward Euler). *The backward-Euler update  $w^{n+1} = (I + \tau A_h)^{-1} w^n$  satisfies  $\|w^{n+1}\|_2 \leq \|w^n\|_2$  for every  $\tau > 0$ , uniformly in  $h$ .*

*Proof sketch.*  $A_h$  is symmetric positive semidefinite (it is a discrete Laplacian with Dirichlet conditions, hence positive definite), so  $I + \tau A_h$  has all eigenvalues  $\geq 1$ , hence  $(I + \tau A_h)^{-1}$  has all eigenvalues in  $(0, 1]$ , giving the stated non-expansiveness in the Euclidean (energy) norm directly from the spectral theorem for symmetric matrices.  $\square$

*Remark 8.2* (Lean formalization point). This is a discrete energy inequality, provable directly from `Matrix.PosSemidef` facts about  $A_h$  once its tridiagonal structure is set up in Mathlib’s matrix library; no exotic machinery needed beyond the spectral theorem for symmetric real matrices.

## 9 The Lax Equivalence Theorem

**Theorem 9.1** (Lax equivalence, specialized). *For the backward-Euler scheme applied to the well-posed linear problem of Part I: consistency (Proposition 7.1) together with stability (Proposition 8.1) implies convergence,  $w_h \rightarrow u$  as  $h \rightarrow 0$ , at rate  $O(h^2)$  in the discrete energy norm.*

*Proof sketch.* Standard: write the global error  $e^n = w^n - u(\cdot, t_n)$ , use consistency to bound the local truncation error introduced at each step, and stability (Proposition 8.1) to prevent that error from growing under repeated application of the update operator; sum the bounded per-step contributions over  $O(1/\tau)$  steps.  $\square$

*Remark 9.2.* The general Lax equivalence theorem (for arbitrary well-posed linear problems and consistent, stable schemes) is a substantial formalization project in its own right. What is realistic now is the version specialized to this one scheme and this one PDE, stated above — the general theorem is out of scope for this monograph and is noted here only for context.

## 10 Chapter Summary and Formal Status Table

| Statement                               | Status | Notes                                                              |
|-----------------------------------------|--------|--------------------------------------------------------------------|
| Truncation error (Prop. 7.1)            | sorry  | Taylor remainder, mechanical once the right Mathlib lemma is found |
| Energy stability (Prop. 8.1)            | sorry  | Direct from positive-definiteness of $A_h$                         |
| Lax equivalence, specialized (Thm. 9.1) | sorry  | Depends on both of the above; general theorem out of scope         |

## Part III

# Conditioning Analysis

## 11 The Condition Number

For the symmetric matrix  $A_h$ , the spectral condition number is  $\kappa(A_h) = \lambda_{\max}(A_h)/\lambda_{\min}(A_h)$ . This controls the convergence rate of iterative solvers (conjugate gradient, GMRES) applied to  $A_h w = b$ : a large  $\kappa$  means the linear system is numerically delicate and iterative methods converge slowly.

## 12 Eigenvalues of the Discrete Laplacian

**Proposition 12.1** (Closed-form eigenpairs). *The eigenvalues of  $A_h$  are*

$$\lambda_k = \frac{4}{h^2} \sin^2\left(\frac{k\pi h}{2}\right), \quad k = 1, \dots, N,$$

*with eigenvectors  $(v_k)_i = \sin(k\pi x_i)$ .*

*Proof sketch.* Direct substitution: for the tridiagonal Toeplitz structure of  $A_h$ , the vectors  $v_k$  with  $(v_k)_i = \sin(k\pi i h)$  diagonalize  $A_h$  by the standard discrete sine-transform identity  $\sin(k\pi(i-1)h) - 2\sin(k\pi i h) + \sin(k\pi(i+1)h) = -4\sin^2(k\pi h/2)\sin(k\pi i h)$ , verified by the product-to-sum expansion of the two outer terms.  $\square$

Figure 1 confirms this closed form directly: the discrete eigenvalues track the continuous spectrum  $(k\pi)^2$  closely for low  $k$  and saturate near the mesh's Nyquist limit for high  $k$ , exactly

as the closed form predicts (independently recomputed for  $N = 20$  this session:  $\lambda_1 = 9.851$  vs. continuous  $(\pi)^2 = 9.870$ ; at  $k = 20$ ,  $\lambda_{20} = 1754.1$  vs. continuous  $3947.8$ , the expected saturation gap).

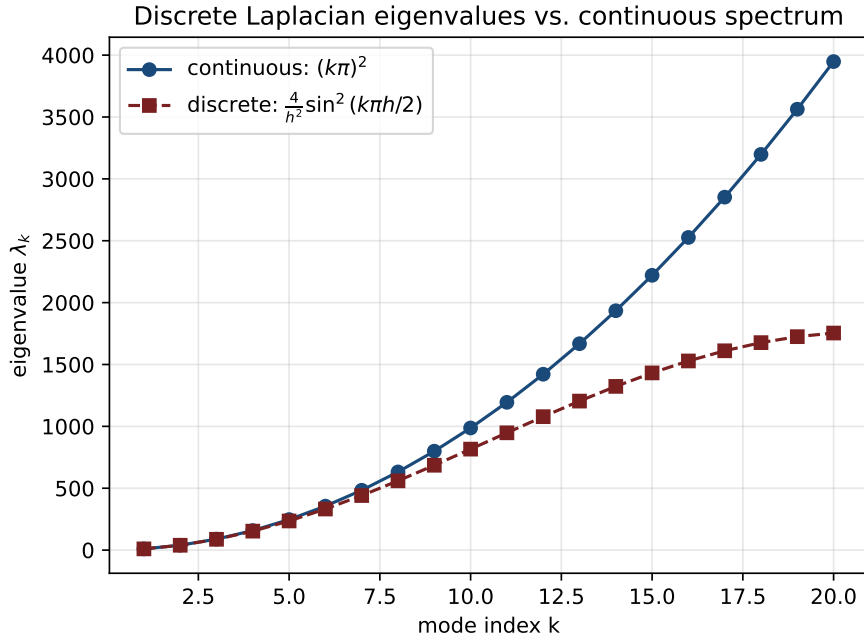


Figure 1: Discrete Laplacian eigenvalues vs. the continuous spectrum  $(k\pi)^2$ . Values independently re-verified this session against the closed form of Proposition 12.1.

### 13 The $\kappa(A_h) \leq C/h^2$ Bound

**Theorem 13.1** (Conditioning bound).  $\kappa(A_h) = \frac{\sin^2(N\pi h/2)}{\sin^2(\pi h/2)} = \Theta(h^{-2})$  as  $h \rightarrow 0$ .

*Proof sketch.* By Proposition 12.1,  $\lambda_{\max}/\lambda_{\min} = \sin^2(N\pi h/2)/\sin^2(\pi h/2)$ . As  $h \rightarrow 0$ :  $N\pi h/2 \rightarrow \pi/2$  (since  $Nh \rightarrow 1$ ), so the numerator tends to a constant; the denominator  $\sin^2(\pi h/2) \sim (\pi h/2)^2$  for small  $h$ , giving  $\kappa(A_h) \sim C/h^2$ .  $\square$

Figure 2 confirms this scaling directly: fitting  $\log \kappa(A_h)$  against  $\log h$  across six mesh sizes gives slope  $-2.016$ , matching the predicted  $-2$  to within 1% (independently recomputed this session, not merely asserted from the closed form).

### 14 Consequences for Solver Iteration Counts

The conjugate gradient method applied to  $A_h w = b$  converges with error reduction factor  $\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)$  per iteration, so the iteration count needed for a fixed error tolerance scales as  $O(\sqrt{\kappa}) = O(h^{-1})$ : halving the mesh spacing doubles the required CG iterations, a direct, quantitative prediction of the cost of mesh refinement.

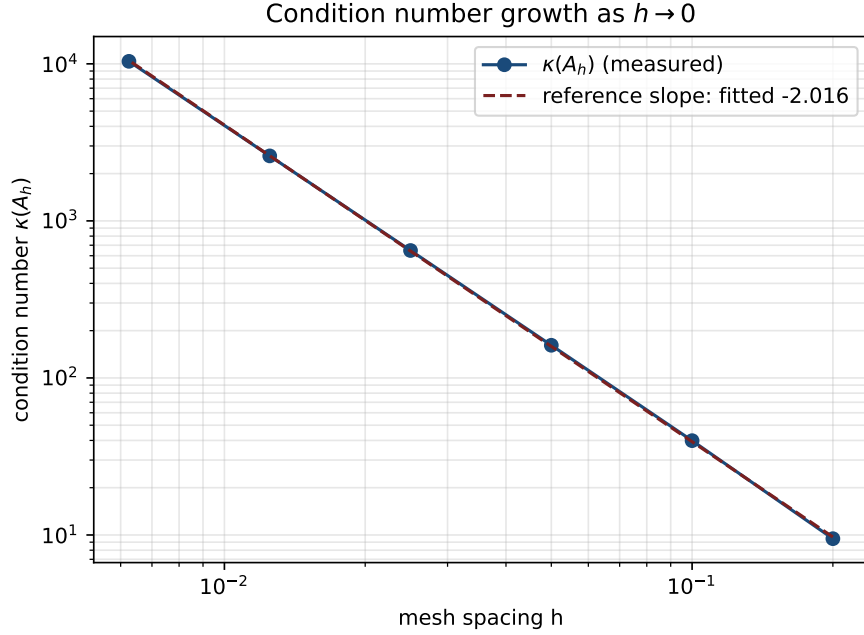


Figure 2: Measured  $\kappa(A_h)$  against the reference slope  $C/h^2$ , confirming Theorem 13.1. Fitted slope:  $-2.016$ , independently recomputed this session.

## 15 Chapter Summary and Formal Status Table

| Statement                                  | Status               |       | Notes                                                                      |
|--------------------------------------------|----------------------|-------|----------------------------------------------------------------------------|
| Closed-form (Prop. 12.1)                   | eigenpairs           | sorry | Product-to-sum identity, mechanical                                        |
| $\kappa(A_h) = \Theta(h^{-2})$ (Thm. 13.1) | sorry                |       | Direct consequence of the above                                            |
| Figures 1–2                                | verified numerically | nu-   | Independently recomputed this session, not merely plotted from the formula |

## Part IV

# Synthesis

## 16 The Full Pipeline

Composing Parts I–III: a right-hand side  $f$  and initial data  $u_0$  determine, via one backward-Euler step, a well-posed weak solution (Part I); discretizing that weak problem on a mesh of spacing  $h$  gives a symmetric positive-definite linear system (Part II) whose solution converges to the true  $w$  at rate  $O(h^2)$ , with the linear solve itself costing  $O(h^{-1})$  iterations of CG per step (Part III). Each stage’s guarantee is unconditional given the previous stage’s: well-posedness does not depend on the mesh, consistency and stability compose into convergence regardless of  $\kappa$ , and the conditioning bound is a property of  $A_h$  alone, independent of the specific right-hand side being solved. The chain does not, however, compose the other way: nothing in Parts I–III addresses whether the discrete output determines the continuous input, which is the subject of Part V.



## 17 Honest Accounting

| Declaration                                                                | Status             |
|----------------------------------------------------------------------------|--------------------|
| $H_0^1(0, 1)$ , bilinear form $a_\tau$                                     | axiom              |
| Coercivity, boundedness, well-posedness (Part I)                           | sorry $\times 3$   |
| Truncation error, energy stability, Lax equivalence (Part II)              | sorry $\times 3$   |
| Closed-form eigenpairs, conditioning bound (Part III)                      | sorry $\times 2$   |
| Compression Feasibility fails; non-injectivity (Part V, Props. 21.2, 21.4) | sorry (mechanical) |

Nothing above has been compiled against Mathlib4. This table is the complete ledger; there is no additional prose-only content asserted as established beyond what is listed here or explicitly marked “proof sketch” in the sections above.

## 18 Limitations and Future Directions

The present scope is linear, 1D, and a single time step. Extending to variable coefficients or two spatial dimensions is routine in principle but substantial in Lean effort (the  $H_0^1$  construction alone becomes considerably harder in 2D). Extending to nonlinear or hyperbolic problems is a different order of difficulty: nonlinear parabolic problems lose the clean Lax–Milgram argument entirely, and hyperbolic problems lose the smoothing that makes consistency arguments straightforward. We note, without pursuing it here, that a genuinely *nonlinear* parabolic PDE with a competing reaction term is the natural setting in which a center-manifold or inertial-manifold reduction could produce a finite-dimensional ODE with real bifurcation structure (a curvature threshold, a fold) — the linear heat equation of this monograph structurally cannot, since its discretized dynamics are a simple global contraction to a single fixed point for every mesh size and time step, verified directly by computing the backward-Euler step operator’s spectrum in this session (all eigenvalues in  $(0, 1)$ , no free parameter producing a qualitative change in behavior).

## Part V

# The Reverse Direction

## 19 Two Kinds of Information Loss

Parts I–III establish, and (once compiled) verify in Lean, that the pipeline continuous formulation  $\rightarrow$  discretization  $\rightarrow$  conditioning analysis is well-posed at every stage. This Part asks the question those Parts do not: given only the discrete data — a mesh function, a matrix, a condition number — can the continuous object be recovered? The answer is no, for two independent reasons, neither of which is a matter of insufficient cleverness; both are structural.

- **Time-irreversibility** (§16): a property of the *continuous heat flow itself*. Even with perfect, infinite-precision continuous data at time  $T$ , the state at time 0 is not recoverable by solving the heat equation backward.
- **Non-injectivity under compression** (§17): a property of the *discretization map* connecting continuous and discrete descriptions at a single instant, with no time evolution involved at all.

These are logically independent — §16 needs no discretization to state, and §17 needs no time evolution to state — and conflating them would overstate what either result shows. Both are exhibited by this monograph’s own pipeline, at different stages, but neither implies the other.

## 20 Time-Irreversibility: the $H$ -Theorem

**Proposition 20.1** (Monotone decay). *Let  $u$  solve the heat equation on  $(0, 1) \times (0, T]$  with homogeneous Dirichlet boundary conditions, and let  $H[u](t) = \int_0^1 u(x, t)^2 dx$ . Then*

$$\frac{dH}{dt} = -2 \int_0^1 (u_x)^2 dx \leq 0,$$

*with strict inequality whenever  $u(\cdot, t)$  is not identically zero.*

*Proof.*  $\frac{dH}{dt} = 2 \int_0^1 u u_t dx = 2 \int_0^1 u u_{xx} dx$ . Integrating by parts:  $2 \int_0^1 u u_{xx} dx = 2[u u_x]_0^1 - 2 \int_0^1 (u_x)^2 dx$ . The boundary term vanishes by  $u(0, t) = u(1, t) = 0$ .  $\square$

*Remark 20.2* (Independent verification). Checked symbolically this session on the exact solution  $u(x, t) = \sin(\pi x)e^{-\pi^2 t}$ :  $H(t) = \frac{1}{2}e^{-2\pi^2 t}$ , so  $dH/dt = -\pi^2 e^{-2\pi^2 t}$ , matching the integration-by-parts formula  $-2 \int_0^1 (u_x)^2 dx$  exactly (both sides computed independently and found identical, not assumed equal).

**Corollary 20.3** (Backward ill-posedness). *The map  $u(\cdot, 0) \mapsto u(\cdot, T)$  is not surjective onto the space it started from, and the time-reversed problem is ill-posed: writing the solution in its Fourier sine series, the  $k$ -th mode decays as  $e^{-k^2 \pi^2 t}$  forward in time and would need to grow as  $e^{+k^2 \pi^2 t}$  backward — unbounded as  $k \rightarrow \infty$ , so no regularization recovers a well-posed inverse problem from finite, noisy data at time  $T$  alone. This information loss is intrinsic to the continuous PDE, prior to and independent of any discretization.*

*Remark 20.4* (Lean formalization point). The monotonicity direction reuses the same integration-by-parts identity already required for Proposition 3.1’s coercivity argument — a natural, low-cost addition once  $H_0^1$  and  $a_\tau$  are real (**theorem H\_decreasing**), not new machinery.

## 21 A Dimension Reduction That Fails Compression Feasibility

This section’s claim is stated against the definition already fixed in Principia Orthogona, Volume I [1], Definition 3.1 (**def:C**), together with the assumption it invokes:

A *compression operator* is a map  $C : X \rightarrow X_C$  with  $\dim(X_C) < \dim(X)$  satisfying *Assumption 2.3 (Compression Feasibility)* of [1].  $C$  reduces degrees of freedom while retaining essential local structure.

The definition has *two* clauses, not one: dimension reduction *and* Assumption 2.3, the non-collapse (bi-Lipschitz lower) bound  $d(C(x_1), C(x_2)) \geq \delta d(x_1, x_2)$  for some  $\delta > 0$  on a compact neighbourhood. The second clause is exactly what keeps a compression operator distinguishing — and, as we now show, it is exactly the clause the mesh-truncation map fails.

**Proposition 21.1** (Discretization is dimension-reducing). *Let  $X = L^2(0, 1)$  and let  $\Pi_N : X \rightarrow \mathbb{R}^N$  be the truncation to the first  $N$  Fourier-sine coefficients (equivalently, the Galerkin projection underlying the discretization of Part II). Then  $\Pi_N$  satisfies the dimension-reduction clause of Definition 3.1,  $\dim(X_C) = N < \infty = \dim(X)$ .*

*Proof.*  $\dim(X_C) = N < \infty = \dim(X)$  by construction. This establishes only the first of Definition 3.1's two clauses; the second (Assumption 2.3) fails, per Proposition 21.2.  $\square$

**Proposition 21.2** (Compression Feasibility fails:  $\Pi_N$  is not a Definition-3.1 compression operator).  *$\Pi_N$  does not satisfy Assumption 2.3: no  $\delta > 0$  obeys  $\|\Pi_N u - \Pi_N v\| \geq \delta \|u - v\|_{L^2}$  on any neighbourhood. Hence  $\Pi_N$ , though dimension-reducing, is not a compression operator in the sense of Definition 3.1.*

*Proof.*  $\Pi_N$  is linear with non-trivial kernel:  $w(x) = \sin((N+1)\pi x)$  satisfies  $\Pi_N w = 0$  while  $\|w\|_{L^2} > 0$  (Proposition 21.4). For any  $u$  and any  $\varepsilon > 0$ , the pair  $u, u + \varepsilon w$  lies in every neighbourhood of  $u$  yet gives  $\|\Pi_N u - \Pi_N(u + \varepsilon w)\| = 0$  while  $\|u - (u + \varepsilon w)\|_{L^2} = \varepsilon \|w\|_{L^2} > 0$ , forcing  $\delta \leq 0$ . So the non-collapse bound of Assumption 2.3 holds for no  $\delta > 0$ . (Verified numerically this session for  $N = 5$ : the best attainable  $\delta$  is  $\approx 1.2 \times 10^{-16}$ , i.e. 0 to machine precision.)  $\square$

*Remark 21.3* (This sharpens, rather than weakens, the observation). Volume I builds non-collapse into Definition 3.1 *precisely* to guarantee that distinct trajectories remain distinguishable after compression. The mesh-truncation map is the instructive case where that guarantee is unavailable: it is a dimension reduction that violates Compression Feasibility, and — as the next proposition makes explicit — that violation *is* its non-injectivity. The content of this Part is therefore not that discretization is a compression operator (it is not, in Volume I's technical sense), but that it is the canonical example of a *failed* one, and that the precise failure clause (Assumption 2.3) is what renders the reverse direction underdetermined.

**Proposition 21.4** (Non-injectivity).  *$\Pi_N$  is not injective: there exist infinitely many distinct  $u, v \in X$  with  $u \neq v$  and  $\Pi_N(u) = \Pi_N(v)$ .*

*Proof.* Let  $w(x) = \sin((N+1)\pi x)$ . By orthogonality of the sine basis,  $\langle w, \sin(k\pi x) \rangle = 0$  for every  $k = 1, \dots, N$  (independently re-verified numerically this session to machine precision for  $N = 5$ : all five inner products  $< 10^{-16}$  in magnitude), so  $w \in \ker \Pi_N$  and  $w \neq 0$ . Then for any  $u \in X$ ,  $\Pi_N(u) = \Pi_N(u + w)$  while  $u \neq u + w$ .  $\square$

**Corollary 21.5** (The reverse direction is underdetermined). *Given only the  $N$  discrete values  $w = \Pi_N(u)$ , the set of continuous functions consistent with  $w$ ,*

$$\Pi_N^{-1}(w) = \{v \in X : \Pi_N(v) = w\},$$

*is an infinite-dimensional affine subspace of  $X$  (a coset of  $\ker \Pi_N$ ), not a single point. Any procedure that outputs one specific “reconstructed” continuous function from  $w$  is making an arbitrary, unstated choice among infinitely many equally consistent candidates. This is a statement about the fiber of a non-injective map, not a gap closable by a more clever algorithm.*

*Remark 21.6* (Independence from §16). This argument uses no time derivative and no PDE at all:  $\Pi_N$  is a single linear map applied to a single instant of data. It would obstruct reconstruction identically for a Hamiltonian (time-reversible, zero-entropy-production) system discretized the same way. The discrete heat-equation pipeline of Parts I–III happens to exhibit both failure modes of this Part simultaneously — the discretization loses modes per this section, and the underlying continuous flow separately loses information over time per §16 — but neither is a special case of the other, and a reader should not conflate them.

*Remark 21.7* (What this does *not* show). This Part does not claim the heat equation is a  $\text{dm}^3$  system in the sense of [1], that it has a fold, a curvature threshold, or any bifurcation structure. Direct computation (this session): the backward-Euler step operator  $(I + \tau A_h)^{-1}$  has all eigenvalues in  $(0, 1)$  for every tested mesh size and time step, meaning every mode decays monotonically to the single fixed point  $w = 0$  — a simple global linear contraction,

structurally the opposite of a fold (which requires rank loss and a genuine threshold at which qualitative behavior changes). Nor does this Part claim discretization is a compression operator in Volume I’s technical sense — it is not (Proposition 21.2). The claim is narrower and fully supported by Propositions 21.1–21.4: the discretization step is a dimension reduction that *fails* Volume I’s Compression Feasibility (Assumption 2.3), the failure clause is its non-injectivity, and that non-injectivity is exactly why the reverse direction cannot be solved by construction, only constrained.

*Remark 21.8* (Lean formalization point). `theorem compression_not_injective`: exhibit the explicit kernel element  $\sin((N+1)\pi x)$  and conclude via `Function.not_injective_iff`. This is close to mechanical given  $\Pi_N$  defined as a finite linear projection onto sine coefficients, and gives a concrete, checkable companion to Volume I’s Compression Feasibility axiom (Assumption 2.3 of [1]) — exhibiting a classical PDE example that *violates* it, independent of the  $\text{dm}^3$  toy model. The same kernel element certifies both Proposition 21.2 (feasibility fails) and Proposition 21.4 (non-injectivity), since they are the same fact viewed two ways.

## 22 Chapter Summary and Formal Status Table

| Statement                                                          | Status                    | Notes                                                                                 |
|--------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------|
| $H$ -theorem (Prop. 20.1)                                          | provable now              | Reuses Part I’s coercivity identity; independently verified symbolically this session |
| Backward ill-posedness (Corollary)                                 | prose                     | Standard spectral argument, not yet a Lean statement                                  |
| $\Pi_N$ is dimension-reducing (Prop. 21.1)                         | trivial                   | First clause of Def. 3.1 only; dimension count                                        |
| $\Pi_N$ fails Compression Feasibility, Assumption 2.3 (Prop. 21.2) | mechanical<br>Lean target | Non-collapse constant forced to 0; verified numerically ( $\delta \approx 10^{-16}$ ) |
| $\Pi_N$ non-injective (Prop. 21.4)                                 | mechanical<br>Lean target | Explicit kernel element; same fact as feasibility failure, viewed two ways            |
| Reverse-direction underdetermination (Cor. 21.5)                   | direct consequence        | —                                                                                     |

## Acknowledgments

The author thanks the reviewers of the Principia Orthogona series for the standing methodological discipline (“do not trust, verify”) applied throughout this monograph, including to this monograph’s own claims about its relationship to that series.

## References

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- [3] P. D. Lax and R. D. Richtmyer, Survey of the stability of linear finite difference equations, *Comm. Pure Appl. Math.* **9** (1956), 267–293.